

CASE

$X_1, X_2, \dots, X_n$ i.i.d.
 $E[X_i] = \mu$ $\text{Var}(X_i) = \sigma^2$



θ
 μ

$\hat{\theta}$
 $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$
 SE: $\sigma/\sqrt{n}$

$\hat{\theta}$

$\bar{x}$

se: standard error estimate

$\frac{s}{\sqrt{n}}$

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

$$ci_{1-\alpha}(\mu) = \bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

confidence interval for μ
 with known variance σ^2

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

$$ci_{1-\alpha}(\mu) = \bar{x} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$$

confidence interval for μ
 with unknown variance

Ⓟ L1. mean (x, sigma, conf.level)

CASE

θ

$\hat{\theta}$

$\hat{\theta}$

se: standard error estimate

$X_1, \dots, X_n$ i.i.d.

2

$X_i \sim \text{Bernoulli}(p)$

$E[X_i] = p, \text{Var}(X_i) = p(1-p)$

p

$\hat{P} \sim N(p, \frac{p(1-p)}{n})$

SE: $\sqrt{\frac{p(1-p)}{n}}$

$\hat{p}$

$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

$\frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \sim N(0,1)$

$\sim N(0,1)$

$ci_{1-\alpha}(p) = \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

confidence interval for p
(variance always unknown)

⑤ $ci(x, \text{success}, \text{conf.level})$

CASE

 θ
 $\hat{\theta}$
 $\hat{\theta}$

se: standard error estimate

 $X_1, \dots, X_{n_x}$ i.i.d.

 $E[X_i] = \mu_x$ $\text{Var}(X_i) = \sigma_x^2$
 $Y_1, \dots, Y_{n_y}$ i.i.d.

 $E[Y_i] = \mu_y$ $\text{Var}(Y_i) = \sigma_y^2$
 x, y independent

$$s_{\text{pool}}^2 = \frac{(n_x - 1)s_x^2 + (n_y - 1)s_y^2}{n_x + n_y - 2}$$

 $\mu_x - \mu_y$
(μ_d)

$$\bar{X} - \bar{Y} \sim N(\underbrace{\mu_x - \mu_y}_{=: \mu_d}, \frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y})$$

$$SE: \sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}$$

 $\bar{x} - \bar{y}$

 If $\sigma_x^2 \neq \sigma_y^2$:

$$\sqrt{\frac{s_x^2}{n_x} + \frac{s_y^2}{n_y}}$$

 If $\sigma_x^2 = \sigma_y^2$:

$$\sqrt{\frac{s_{\text{pool}}^2}{n_x} + \frac{s_{\text{pool}}^2}{n_y}}$$

$$\frac{\bar{X} - \bar{Y} - (\mu_x - \mu_y)}{\sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}} \sim N(0,1)$$

 $\sim N(0,1)$

$$\rightarrow CI_{1-\alpha}(\mu_x - \mu_y) = \bar{x} - \bar{y} \pm Z_{\alpha/2} \sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}$$

 confidence interval
for $\mu_x - \mu_y$,
 σ_x^2, σ_y^2 known

$$\frac{\bar{X} - \bar{Y} - (\mu_x - \mu_y)}{\sqrt{\frac{s_{\text{pool}}^2}{n_x} + \frac{s_{\text{pool}}^2}{n_y}}} \sim t_{n_x + n_y - 2}$$

 $\sim t_{n_x + n_y - 2}$

$$\rightarrow CI_{1-\alpha}(\mu_x - \mu_y) = \bar{x} - \bar{y} \pm t_{n_x + n_y - 2} \sqrt{\frac{s_{\text{pool}}^2}{n_x} + \frac{s_{\text{pool}}^2}{n_y}}$$

 confidence interval
for $\mu_x - \mu_y$,
unknown equal
variances $\sigma_x^2 = \sigma_y^2$

(P) LI. diff mean (x, y, type = "independent", sigma.x, sigma.y, conf.level)

CASE

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$\hat{\theta}$

$\hat{\theta}$

se: standard error estimate

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$(x_1, y_1), \dots, (x_n, y_n)$

$D_1, D_2, \dots, D_n$ i.i.d.

$E[D] = \mu_x - \mu_y$

$\text{Var}(D) = \sigma_x^2 + \sigma_y^2 - 2\sigma_{xy}$
 $= \sigma_D^2$

$\mu_x - \mu_y$
 (μ_D)

$\bar{D} = \bar{X} - \bar{Y} \sim N(\mu_x - \mu_y, \frac{\sigma_D^2}{n})$

$\bar{x} - \bar{y}$
 $(\bar{d})$

$\frac{s_d}{\sqrt{n}}$

recall s_d is calculated:
 ① as standard deviation of the differences
 ② with $s_d^2 = s_x^2 + s_y^2 - 2s_{xy}$

X, Y PAIRED

$D_i = X_i - Y_i$

$\frac{\bar{D} - \mu_D}{\sigma_D / \sqrt{n}} \sim N(0, 1)$

$\longrightarrow C_{1-\alpha}(\mu_D) = \bar{d} \pm Z_{\alpha/2} \frac{\sigma_D}{\sqrt{n}}$

confidence interval for $\mu_D (= \mu_x - \mu_y)$, variance σ_D^2 known

$\frac{\bar{D} - \mu_D}{s_D / \sqrt{n}} \sim t_{n-1}$

$\longrightarrow C_{1-\alpha}(\mu_D) = \bar{d} \pm t_{n-1, \alpha/2} \frac{s_D}{\sqrt{n}}$

confidence interval for μ_D , variance unknown

② `ci.diffmean(x, y, type = "paired", sigma.d, conf.level)`

CASES

$X_1, \dots, X_{n_x}$ i.i.d.
 $X_i \sim \text{Bernoulli}(p_x)$
 $Y_1, \dots, Y_{n_y}$ i.i.d.
 $Y_i \sim \text{Bernoulli}(p_y)$
 X, Y independent

J

 θ
 $\hat{\theta}$
 $\hat{\theta}$

se: standard error estimate

 $p_x - p_y$
 $\hat{p}_x - \hat{p}_y$

$$N(p_x - p_y, \frac{p_x(1-p_x)}{n_x} + \frac{p_y(1-p_y)}{n_y})$$

 $\hat{p}_x - \hat{p}_y$

$$\sqrt{\frac{\hat{p}_x(1-\hat{p}_x)}{n_x} + \frac{\hat{p}_y(1-\hat{p}_y)}{n_y}}$$

$$\frac{\hat{p}_x - \hat{p}_y - (p_x - p_y)}{\sqrt{\frac{\hat{p}_x(1-\hat{p}_x)}{n_x} + \frac{\hat{p}_y(1-\hat{p}_y)}{n_y}}}$$

 $\sim N(0,1)$

$$ci_{1-\alpha}(p_x - p_y) = \hat{p}_x - \hat{p}_y \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_x(1-\hat{p}_x)}{n_x} + \frac{\hat{p}_y(1-\hat{p}_y)}{n_y}}$$

Confidence interval for $p_x - p_y$
 (variance always unknown)

(R) `ci.diffprop(x, y, success.x, success.y, conf.level)`